

Global Internet Adoption and the Gender Digital Divide: A World Development Indicators (WDI) Analysis Report

AnalystLab Africa Data Analytics Capstone Project

Week 8: Global Internet Adoption Documentation, Visualization and Presentation

Dataset: The World Development Indicators (WDI)

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Executive Summary

This project analyzes global internet adoption using the World Bank's World Development Indicators (WDI) dataset. The objective is to examine global internet usage trends, compare digital adoption across countries and regions, and assess the gender digital divide in internet access. Using Microsoft Power BI, the dataset was cleaned, transformed, and modeled into an interactive dashboard consisting of three analytical pages: Executive Overview, Country Performance Analysis, and Gender Digital Divide. The analysis highlights regional disparities, identifies leading and low-performing countries, and reveals persistent gender gaps in internet adoption. The findings provide evidence-based insights that can support policymakers, development organizations, and stakeholders in promoting equitable digital inclusion.

Objective / Problem Statement

Internet access has become one of the most important indicators of economic development, education, innovation, and social inclusion. Despite remarkable growth in global internet adoption over the past three decades, significant disparities remain across countries, income groups, regions, and gender.

The objective of this project is to analyze global internet adoption using the World Bank World Development Indicators dataset to identify trends, compare country performance, evaluate regional differences, and assess the gender digital divide. The project aims to transform raw development data into actionable insights through interactive dashboards that support informed decision-making and promote digital inclusion.

Dataset Description

The dataset used for this project was obtained from the World Bank World Development Indicators (WDI) database. WDI is the World Bank's flagship repository of global development statistics covering more than 200 countries and territories.

The dataset includes indicators across multiple development sectors, including economic growth, education, health, infrastructure, governance, technology, trade, and environmental sustainability.

For this project, internet adoption indicators were selected to evaluate digital inclusion across countries, regions, income groups, and gender over time.

Data Cleaning Process

The dataset underwent several preprocessing and transformation steps before analysis:

- Imported the WDI CSV dataset into Power BI.
- Removed irrelevant variables that were not related to internet adoption.
- Filtered the dataset to retain internet usage indicators.
- Checked for missing values and excluded incomplete records where necessary.
- Standardized country and regional names for consistency.
- Converted data types to appropriate formats.
- Removed duplicate records.
- Created calculated measures using DAX for:
 - Average Internet Usage
 - Highest Internet Usage
 - Average Male Internet Usage
 - Average Female Internet Usage
 - Average Gender Gap
 - Countries Analysed
 - Latest Year
 - Lowest Internet Usage
- Established relationships between tables to enable interactive filtering.

These steps improved the accuracy, consistency, and reliability of the analysis.

Analysis Methodology

The analysis followed the complete data analytics workflow:

- Data Collection
- Data Cleaning
- Data Transformation
- Exploratory Data Analysis
- Dashboard Development
- Insight Generation
- Recommendation Development

Microsoft Power BI was used for visualization, DAX calculations, KPI development, and interactive dashboard creation.

Dashboard Overview

Page 1 – Executive Overview

The Executive Overview provides a high-level summary of global internet adoption trends.

Key components include:

- Countries Analysed
- Average Internet Usage
- Highest Internet Usage
- Latest Year (2025)
- Internet Adoption Trend
- Regional Internet Usage Comparison
- Income Group Distribution
- Interactive filters by Year, Country, Region, and Income Group

The page enables stakeholders to quickly understand global internet adoption patterns and regional differences.

Page 2 – Country Performance Analysis

This page compares internet adoption across individual countries.

It includes:

- Average Internet Usage
- Highest Internet Usage
- Lowest Internet Usage
- Countries Analysed
- Top 10 Countries by Internet Usage
- Bottom 10 Countries by Internet Usage
- Summary Insights
- Recommendations
- Interactive filters

The analysis identifies countries leading digital adoption as well as countries requiring greater investment in digital infrastructure.

Page 3 – Gender Digital Divide

The Gender Digital Divide dashboard evaluates disparities in internet access between males and females.

It includes:

- Average Female Usage
- Average Male Usage
- Average Gender Gap
- Highest Female Usage
- Highest Male Usage
- Gender Trend Analysis
- Regional Gender Gap
- Countries with Largest Gender Gap
- Key Insights

This page highlights inequalities in digital access and supports evidence-based strategies for improving gender equity.

Key Findings

- A total of **245 countries and territories** were analyzed.
 - The overall average internet usage across countries is **43.4%**.
 - Internet adoption has increased steadily from **1990** to **2025**.
 - **North America** records the highest average regional internet usage (**62.7%**), followed by Europe & Central Asia (**59.8%**).
 - **Sub-Saharan Africa** has the lowest regional average internet usage (**13.3%**).
 - **Kosovo** recorded the highest average internet usage among countries analyzed (**86.6%**).
 - Several countries, including Niger, South Sudan, and Chad, recorded very low internet adoption.
 - Average male internet usage (**69.0%**) is slightly higher than average female internet usage (**65.6%**).
 - The average global gender gap in internet usage is **3.7%**.
 - **South Asia** and **Sub-Saharan Africa** exhibit the largest regional gender disparities.
 - **Türkiye** recorded the largest country-level gender gap.
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Insights

The analysis indicates that global internet adoption has improved considerably over the last three decades; however, digital inequality remains evident across regions and income groups.

Countries with higher income levels generally demonstrate greater internet penetration due to stronger digital infrastructure, better affordability, and wider technological access. Conversely, many low-income countries continue to experience limited connectivity.

The gender analysis reveals that although the global gender gap is relatively modest, substantial disparities persist in specific regions, particularly South Asia and Sub-Saharan Africa. Closing these gaps will require targeted policies focused on affordability, education, digital literacy, and inclusive infrastructure.

Recommendations

Based on the findings, the following recommendations are proposed:

- Increase investment in broadband infrastructure in underserved regions.
- Expand affordable internet access through public-private partnerships.
- Promote digital literacy programs, particularly for women and girls.
- Develop policies that address gender barriers to digital access.

- Prioritize investments in low-income countries with limited internet penetration.
 - Encourage continuous monitoring of internet adoption indicators to support evidence-based policymaking.
 - Strengthen regional collaboration to reduce inequalities in digital connectivity.
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Conclusion

This project successfully applied the complete data analytics workflow to analyze global internet adoption using the World Development Indicators dataset. Through effective data cleaning, transformation, visualization, and analysis, the project produced an interactive Power BI dashboard that provides valuable insights into global internet usage, regional disparities, country performance, and gender differences.

The findings demonstrate that while internet adoption continues to grow globally, significant inequalities remain across regions, income groups, and gender. Addressing these disparities will require sustained investment, inclusive digital policies, and international collaboration to ensure equitable access to digital technologies for all.